



Vis-Assist: computer vision and haptic feedback-based wearable assistive device for visually impaired

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Abstract

Visual impairment affects millions of people worldwide, posing significant challenges in their daily lives and personal safety. While assistive technologies, both wearable and non-wearable, can help mitigate these challenges, wearable devices offer the advantage of hands-free operation. In this context, we present Vis-Assist, a novel wearable visual assistive device capable of detecting and classifying objects, measuring their distances, and providing real-time haptic feedback through a vibration motor array, all using an integrated low-cost computational unit without the need for external servers. Our study distinguishes itself by utilizing haptic feedback to convey object information, allowing visually impaired individuals to discern between 19 different object classes following a brief training period. Haptic feedback offers an alternative to audio that doesn't block hearing and can be used alongside it, serving as a complementary solution. The performance of the developed wearable device was evaluated through two types of experiments with four participants. The results demonstrate that users can identify the location of objects and thereby prevent collisions with obstacles. The experiments conducted demonstrate that users, on average, can locate a predefined object, such as a chair, within a 40 m² vacant space in under 94 seconds. Furthermore, users exhibit proficiency in finding objects while navigating around obstacles in the same environment, achieving this task in less than 121 seconds on average. The system developed here has high potential to help the self-navigation of visually impaired people and make their daily lives easier. To facilitate further research in this field, the complete source code for this study has been made publicly available on GitHub.

Keywords Wearable Device · Visually Impaired · Assistive Technology · Haptic Feedback · Real-Time Detection · Edge-AI

1 Introduction

There are millions of people in the world who suffer visual impairment [1]. Visual impairment can present challenges in certain daily activities, particularly in mobility and navigation, but many visually impaired individuals lead independent and active lives. Assistive technologies continue to play a vital role in enhancing accessibility and safety. A wide range of assistive technologies, both wearable and non-wearable, have been developed to support navigation and object avoidance for the visually impaired. These devices prevent users from colliding with objects and enable safe

movement. Ensuring the affordability of fabricating these devices is crucial for enhancing accessibility, as highlighted by recent studies on cost-effective design approaches [2].

The present study introduces a novel wearable device Vis-Assist capable of object detection and classification by implementing the YOLOv7 (You Only Look Once) model, facilitated by the cost-effective computational unit, Jetson Nano. Additionally, the device integrates an array of vibration motors, to inform the user regarding the presence of objects in their proximity and provide information concerning their respective distances from the user. Although audio-based solutions are widely used in assistive technologies, there are scenarios where haptic feedback could be beneficial, such as when a user is engaged in conversation or relying on auditory cues for navigation. Instead of positioning haptic feedback as a superior alternative, we consider it a complementary approach that enhances accessibility by providing additional sensory input. In contrast to prior investigations documented in the extant literature, our study employs an array of vibra-

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tion motors to impart substantial information through haptic feedback to the user. This innovative approach has enabled the proficient acquisition of discernible distinctions among 19 distinct object classes subsequent to a brief training period.

The remainder of this paper is structured as follows. Sect 2 presents a comprehensive review of relevant literature. In Section 3, the development methodology of the proposed device is detailed. Sect 3.1 outlines the hardware components of the wearable system, while Sect 3.2 elaborates on the machine learning algorithm employed for object detection and classification. Distance measurement using 1D lidar is described in Sect 3.3, and the feedback methodology, including the vibration motor array and associated patterns, is presented in Sect 3.4. Experimental methodologies, results, and discussion are provided in Sect 4. Finally, concluding remarks are offered in Sect 5.

2 Literature review

In the literature, there are various studies on developing wearable devices for visually impaired people. The types of sensors used, the methods of feedback given to the user, and the richness of the information given to the user vary in these studies. Studies in the literature can be divided into two Haptic Feedback and Audible Feedback according to their feedback methods. The feedback methods can affect the response time, learnability, and richness of the information. Audible feedback offers learnability, fast response time, and rich information, but the richness of the information and fast response time in haptic feedback affect learnability negatively. The most advantageous part of haptic feedback is that it doesn't replace any of the sensing organs of the human; it adds a new sensing mechanism based on another. In studies that give audible feedback, although the information provided to the user can be rich and faster, hearing, the best natural navigation sense of visually impaired individuals, can be blocked. While audio-based solutions can interfere with hearing, technologies such as bone conduction headphones and 'pass-through' Bluetooth earphones help mitigate this issue by allowing users to perceive both audio feedback and environmental sounds. However, there are still situations where haptic feedback may be preferable, such as during conversations or when listening to other sounds.

Haptic feedback wearable devices use different types of stimulators, such as vibration motors or electrodes on the skin to inform the user about the environment or navigate the user to reach the desired destination. In the literature, haptic feedback wearable devices for the visually impaired mostly inform the user about the distance. Prattico et al. propose a system that informs the user with vibration motors when any obstacle is detected up to 6 meters range by using infrared and acoustic distance sensors [3]. Katzschmann et

al. and Cardin et al. propose using multiple distance sensors and multiple vibration motors corresponding to each of them to inform the user about the distances of the obstacles with increased field of view [4], [5]. Brown et al. propose using a 3D camera instead of multiple distance sensors, and they divide the field of view of the camera into six pieces and use six vibration motors to inform the user about distances up to 2 meters range for six directions [6]. Bahadir et al. utilize multiple vibration motors and distance sensors in their design. However, instead of informing the user about distances to obstacles, within 6 meters range, their system employs vibration motors to guide users by indicating when to turn right or left, and by what angle [7]. Petsiuk et al. propose low-cost bracelet-type devices that contain one vibration motor and one ultrasonic distance sensor [8]. Lee et al. propose a more complex device which is able to create a map of the environment by using an RGB-D (Red-Green-Blue-Depth) camera and SLAM (Simultaneous localization and mapping) and it navigates the user by vibration motors [9]. Velázquez et al. and Tachiquin et al. propose a GPS (Global Positioning System) based outdoor navigation system, which employs four vibration motors positioned inside the shoe. This design assists the user in rotating in four distinct directions, facilitating navigation to reach the desired destination [10, 11]. Lee et al. propose a wearable device that contains multiple vibration motors to navigate the user with commands such as turn right, turn left, stop, and forward to reach the desired destination by using GPS and acoustic distance sensor array. Measured distances were limited to 2 meters; only values of 2 meters or less were taken into account [12]. Fei et al. propose a wearable device that integrates ORB-SLAM2 and YOLOv5. The device consists of a glove equipped with a depth camera and haptic stimulators. The used camera provides distance measurement within 10 meters range. Experiments were conducted to evaluate the device's effectiveness in grasping specific objects on a table, using feedback from the haptic stimulators [13]. Bouteraa et al. propose a wearable device that contains mini-lidar and ultrasonic sensors for distance measurement, and IMU (Inertial Measurement Unit) for calculating orientation and velocity. Their device uses both audio and haptic feedback to inform the user to avoid collision. The performance of the device was evaluated by both totally blind and partially sighted participants using different walking paths in indoor and outdoor environments. The evaluation metric was the number of collisions, and the device was compared to a traditional cane. The results show that the number of collisions was significantly reduced when using the device, dropping from 61 to 12 in total [14]. Ghatkamble et al. present a footwear system incorporating ultrasonic sensors to prevent collisions. They further enhance the system's capabilities through the integration of a camera and laptop for object detection and classification. The proposed system

employs both haptic and auditory feedback mechanisms to convey information to the user [15].

Audio feedback in wearable devices provides visually impaired individuals with comprehensive environmental information, such as object names, distances, and directional guidance. However, this approach may impede hearing, which is the most crucial sensing mechanism for visually impaired people. Audio feedback is more common in studies in the literature due to its ease of application. Budrionis et al. show that smartphone-based devices mostly use audio feedback in their review [16]. Mocanu et al. and Tapu et al. have developed a wearable device that integrates a smartphone as a data acquisition unit, a laptop as the processing unit, and bone-conduction headphones for non-intrusive acoustic feedback. This device performs object detection, classification, and object tracking, and relay information to the user acoustically. In addition, the device they developed includes 4 ultrasonic sensors, which allows for distance measurement of 6 meters in 4 directions [17, 18]. Similarly, Sövény et al. have proposed a wearable system comprising a camera-mounted helmet and a portable computer, designed to detect and identify traffic situations, providing navigation through audio feedback. The device can detect up to 10 meters away. The device is tested in various public settings with a diverse group of users more than 50 people, including visually impaired individuals and children. Most users responded positively, though a few visually impaired participants felt the device was noisy [19]. Takefuji et al. introduced a device using an RGB-D camera alongside an audio feedback mechanism to inform visually impaired people about the environment. The used RGB-D camera provides distance measurement for each pixel up to 20 meters range. They employed the You Only Look Once (YOLO) algorithm to detect and classify the objects. The performance of the proposed solution was evaluated with 69 visually impaired participants. The questionnaire results show that 70 percent of the participants reported they could go out with confidence when using the device [20]. Additionally, Yang et al. developed a wearable device capable of semantic segmentation, navigating users with audio cues to identify walkable paths and informing them about various object types such as persons, vehicles, and stairs [21]. Hsieh et al. aim to provide the direction of the safe route by audio feedback to the visually impaired user. In their work, an RGB camera is used and the depth map is calculated by the CNN (Convolutional neural network) method. The feedback to the user contains three commands: left, center, right [22]. Mohanraj et al. introduce a wearable assistive device comprising two distinct components. The first component is a head-mounted unit equipped with a Raspberry Pi and camera. The second component is a white cane integrated with ultrasonic and water sensors. The system operates independently on each component, providing auditory feedback to the user. Environmental data gathered includes detected

object categories, proximity to obstacles within 4 meters range, and the presence of wet surfaces [23]. Maya-Martínez et al. propose a machine learning model architecture that enables object detection and classification on low-cost edge devices. The experiments demonstrate that Tiny-YOLOv3 can run on the Jetson Nano with an acceptable processing speed [24]. The overall comparison of the reviewed studies are shown in Table 1.

The literature review identifies a gap in current technologies, with no devices solely using haptic feedback to inform visually impaired users about objects. While auditory feedback is common, it can interfere with hearing. Bone conduction and pass-through earphones reduce this by preserving environmental awareness. Still, haptic feedback can be better during conversations or when listening to other sounds. Haptic feedback, though less utilized, adds an extra sensory input without disrupting other senses, highlighting the need for innovative solutions that enhance object recognition for the visually impaired.

3 Methods

Visually impaired people need help to move around safely and independently. Their main needs include avoiding obstacles, recognizing objects, understanding directions, and staying aware of their surroundings without blocking their hearing. The primary objective of this study is to design and create a wearable device, named Vis-Assist, with the capability to acquire visual input and distance measurements, subsequently relaying this information to the user through haptic feedback in the form of object categorization and distance estimation. The process involves the integration of machine learning algorithms to discern and categorize objects within the camera frames. The flow diagram of the proposed system is shown in Fig. 1. Notably, all computational procedures are executed within the device itself, eliminating the necessity for external connections to remote systems. It is important to underline that this solution does not include commands to direct the user in a certain direction; rather, it aligns with the navigation approach employed by individuals without impairments, who predominantly rely on visual observation of their surroundings to make informed decisions regarding their actions.

3.1 Device design

The device contains three main parts, which are connected with wires. These are the sensor box, controller box, and feedback system. The visual representations and the positions of these parts on the user are shown in Figure 2. Figure 3 shows the real state of all parts of the device together.

Table 1 Summary of recent wearable assistive devices and systems for visually impaired individuals. The table highlights key characteristics of each study, including the data acquisition methods, feedback type, main advantages, and limitations

Authors	Sensors	Interface	Advantages	Limitations
Prattico et al. (2013) [3]	Infrared and acoustic sensors	Haptic	Low-Cost	Limited to obstacle detection
Katzschmann et al. (2018) [4]	Distance sensors	Haptic	Wider field of view, Low-Cost	Limited to obstacle detection
Cardin et al. (2007) [5]	Distance sensors	Haptic	Wider field of view, Low-Cost	Limited to obstacle detection
Brown et al. (2019) [6]	3D camera	Haptic	Divides view into 6 directions, 2m range	Short range (2 meters), Limited to obstacle detection
Bahadir et al. (2012) [7]	Distance sensors	Haptic	Provides navigation commands	Limited to obstacle detection
Petsiuk et al. (2019) [8]	Ultrasonic sensor	Haptic	Low-cost bracelet design	Limited to obstacle detection
Lee et al. (2016) [9]	RGB-D camera	Haptic	Can create environmental map	High Cost, Limited to obstacle detection
Velázquez et al. (2018) [10]	GPS	Haptic	Enables directional navigation, Low-Cost	Only outdoor, No environmental information
Tachiquin et al. (2021) [11]	GPS	Haptic	Enables directional navigation, Low-Cost	Only outdoor, No environmental information
Lee et al. (2016) [12]	GPS, acoustic sensor array	Haptic	Navigation with commands, Low-Cost	2 meters range, obstacle detection only
Fei et al. (2024) [13]	RGB-D camera	Haptic	Object grasping tasks supported, 10m range	High Cost, Designed for tabletop scenarios
Bouteraa et al. (2021) [14]	Mini-lidar, ultrasonic, IMU	Haptic + Audio	Fuzzy decision support, Low-Cost	Limited to obstacle detection
Ghatkamble et al. (2023) [15]	Ultrasonic, camera	Haptic + Audio	Object detection + classification	Requires laptop
Mocanu et al. (2016) [17]	Smartphone, ultrasonic sensors	Audio	Object detection, classification	No depth sensing, Requires laptop
Tapu et al. (2017) [18]	Smartphone, ultrasonic sensors	Audio	Tracks objects, 6m range in 4 directions	No depth sensing, Requires laptop
Sóvény et al. (2015) [19]	Camera	Audio	Detection of crosswalks, stairs, hanging and ground obstacles	Requires computer in backpack
Takefuji et al. (2020) [20]	RGB-D camera	Audio	Detects objects up to 20m. 70% positive user feedback	High Cost, relies on cloud
Yang et al. (2018) [21]	Camera	Audio	Semantic segmentation and object type feedback	High Cost
Hsieh et al. (2020) [22]	Camera	Audio	Safe route guidance with 3 commands	High Cost
Mohanraj et al. (2024) [23]	Camera, ultrasonic, water sensor	Audio	Detects proximity, water, object type	Limited evaluation

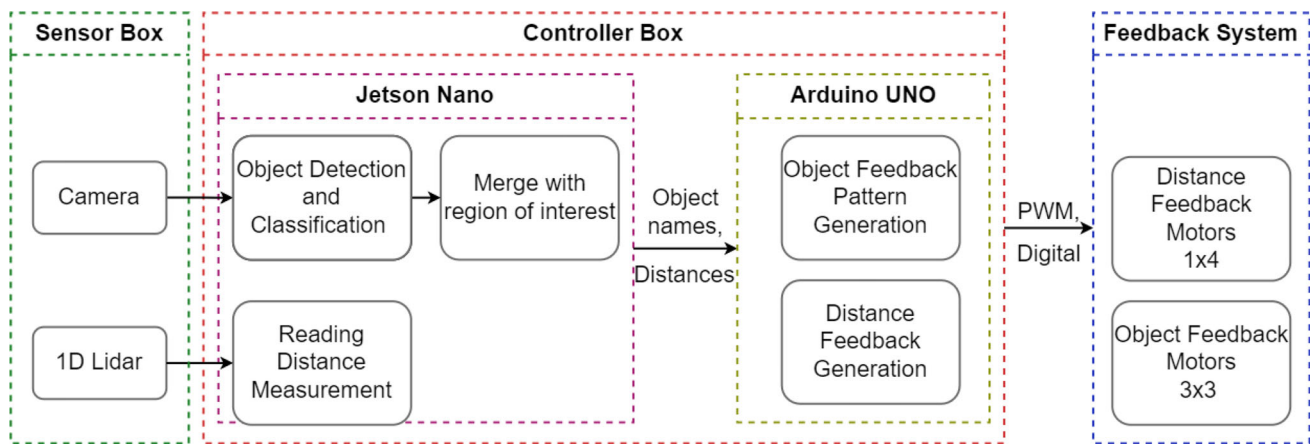


Fig. 1 Flow diagram of the Vis-Assist system for visually impaired users. The device captures environmental data through a camera and LiDAR sensor, processes it for object detection and distance measure-

ment, and conveys this information to the user via two distinct haptic feedback arrays, providing non-auditory environmental awareness

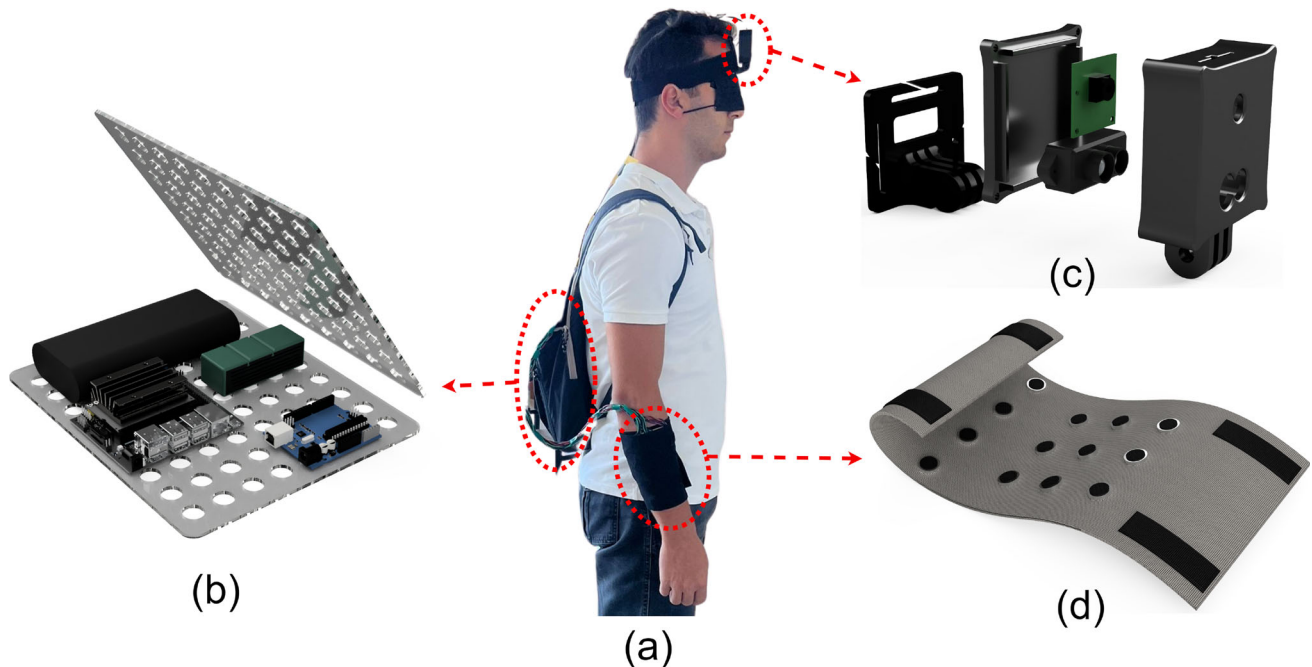


Fig. 2 The proposed wearable Vis-Assist device for the visually impaired to provide environmental information such as detected objects and corresponding distances to the user via haptic feedback, (a) The user wearing the device, (b) The controller box, (c) The sensor box, (d) The haptic feedback band

The sensor box accommodates a pair of distinct sensors, namely a distance sensor and a visual sensor, both of which serve to capture environmental stimuli. This enclosure is affixed to a headband apparatus strategically positioned to facilitate autonomous environmental surveillance by the user. Notably, the sensor unit exclusively undertakes data acquisition, devoid of any embedded processing or power capabilities. Subsequently, the acquired data are relayed to the central controller unit for further analysis and action.

The controller box encompasses an Nvidia Jetson Nano employed as the principal processing unit, and an Arduino UNO serving as the haptic feedback controller.

Additionally, the configuration incorporates dedicated battery and vibration motor drivers. The Nvidia Jetson Nano, recognized for its cost-effectiveness, functions as a proficient processing unit with the capability to execute machine learning algorithms leveraging its integrated GPU (Graphics processing unit) processing capabilities. The physical placement of the controller box can be adapted to either a backpack

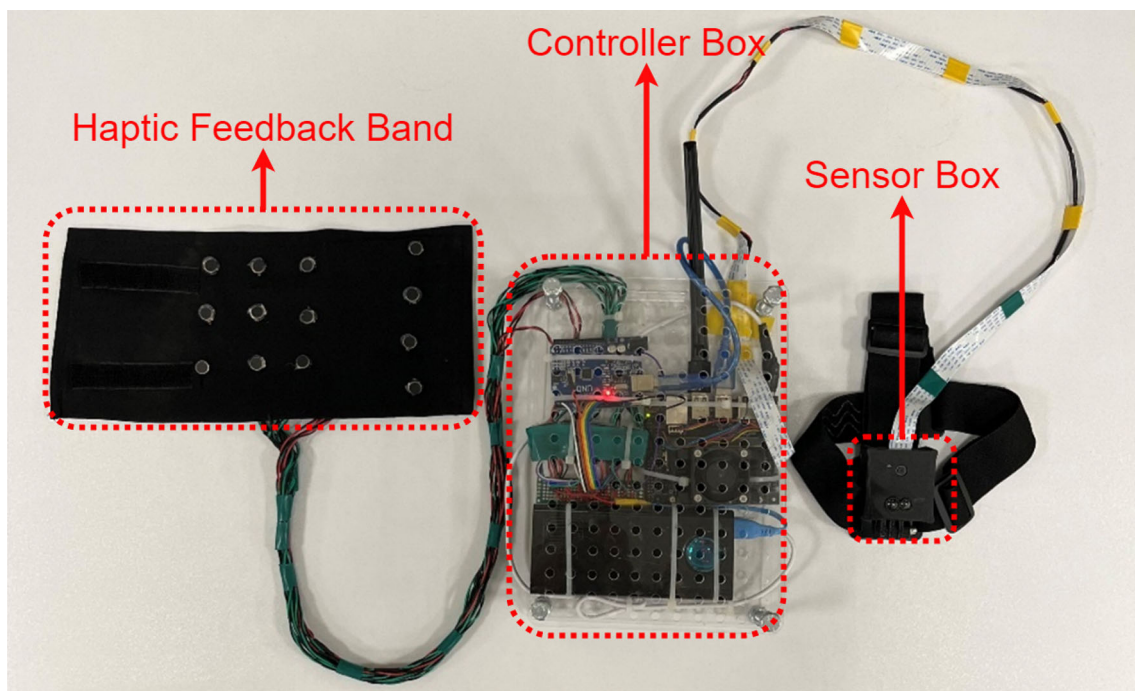


Fig. 3 Components of the Vis-Assist system. The image displays the head-mounted sensor box containing camera and LiDAR, controller box housing Nvidia Jetson Nano and Arduino UNO, and haptic feedback armband with vibration motor arrays for object and distance information

configuration or a chest band arrangement, providing flexibility in usability and user experience.

In the controller box, a rechargeable battery is integrated, boasting a power capacity of 65 watts. Notably, all constituent elements of the wearable device collectively draw a power consumption of 10 watts. This observation establishes that the wearable device can operate for a minimum duration of 6.5 hours on a single charge of its battery source.

The implemented feedback system comprises a 3x3 array of vibration motors designed to generate discernible patterns corresponding to distinct object classes, coupled with a 1x4 vibration motor array dedicated to conveying precise distance-related information to the user. These vibratory motor arrays are strategically integrated into an armband, ensuring seamless wearability and user comfort.

3.2 Object detection and recognition model

YOLO, an acronym for You Only Look Once, is a notable object detection algorithm in computer vision. It distinguishes itself through its single-pass processing of an entire image using a convolutional neural network. This approach allows for real-time object detection with commendable speed and accuracy. YOLO's network architecture divides the input image into a grid and simultaneously predicts bounding boxes and class probabilities for objects within each grid cell. This approach minimizes computational redundancy and has found application in various domains,

including autonomous navigation, surveillance, and image and video analysis [25].

In this work, YOLOv7-Tiny is used for object detection and classification [26]. Some modifications have been made before using it, such as limiting the object classes to avoid unnecessary object classes being detected and disabling the drawing rectangles around the objects. The only necessary information is the corner coordinates of the detected objects and their classes. Then, the algorithm is developed to check if the region of interest, which is a circle in the center of the camera frame, is in any of the detected object rectangles. If this condition is satisfied, the detected object name needs to be transferred to the user with haptic feedback.

The YOLOv7-tiny algorithm is a derivative of YOLOv7, incorporating the cascade-based model scaling strategy while enhancing the efficiency of the long-range aggregation network (ELAN) to achieve improved detection accuracy with fewer parameters and quicker detection speed. The architecture of the YOLOv7-Tiny is shown in Fig. 4. The ELAN layer consists of a cascade of CBL layers. Each CBL contains a convolutional layer, batch normalization process, and SiLU activation function. The SiLU (Sigmoid Linear Unit) activation function improves non-linearity and gradient flow, which helps the network learn complex features more effectively. Within the backbone, there are CBL convolutional blocks, an enhanced efficient long-range aggregation network (ELAN) layer, and an MPConv convolutional layer. CBL layers (Convolution-BatchNorm-LeakyReLU) contribute to better

The object classes used in this study were selected from the COCO dataset, which includes a diverse set of 80 object categories. To optimize the system for real-world navigation, we focused on objects that are most relevant for visually impaired users, such as people, furniture, vehicles, and obstacles commonly found in indoor and outdoor environments. Less relevant objects, such as animals, food items, and small household objects, were excluded to prevent unnecessary detections and reduce processing time. This selection ensures that the system provides meaningful feedback without overwhelming the user with irrelevant information.

The acquisition of distance information holds paramount significance in ensuring the autonomous mobility of individuals with by visual impairments. To fulfill this purpose, the TF-mini lidar takes on the role of the distance-measuring sensor. Operating on a wavelength of 850 nm, this lidar system leverages infrared light to execute distance measurements grounded in the time-of-flight principle. Its operational capabilities are enhanced by a heightened sensitivity to light, thereby rendering it suitable for deployment in outdoor environments. Diverging from traditional acoustic sensors, the efficacy of infrared sensors in gauging distances remains independent of object geometry. Additionally, the sensor exhibits remarkable precision, boasting an accuracy level of $\pm 4\text{cm}$ within the range of 0.3m to 6m, and a slightly extended accuracy of $\pm 6\text{cm}$ spanning from 6m to 12m. Notably, the distance measurement process is executed at the sensor apparatus itself, with subsequent transmission of distance values facilitated through a UART connection [28].

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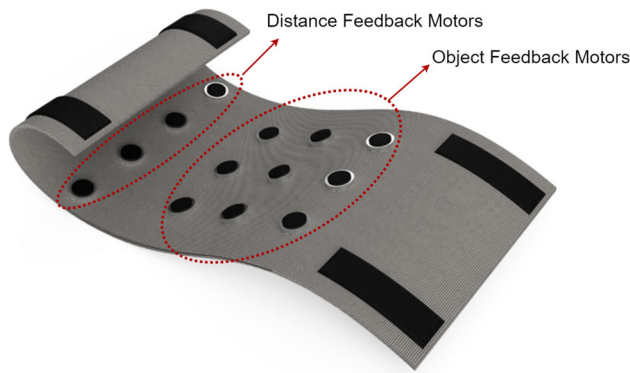


Fig. 5 The illustration of the haptic feedback system. It contains a 1x4 vibration motor array for distance feedback and a 3x3 vibration motor array for object feedback

3.4 Feedback system

The feedback mechanism comprises two distinct arrays of vibration motors. One of these arrays is dedicated to conveying measured distance values to the user through haptic stimuli. Specifically, the distance feedback vibration motor array is structured as 1x4 in size. The second array serves the purpose of relaying information about the detected object class situated at the center of the camera frame. This array, termed the object feedback vibration motor array, is configured in a 3x3 layout. Keeping these arrays physically separate prevents mechanical vibration interference between motors and avoids confusion at the user side, ensuring a more reliable and comfortable experience. The 3 × 3 object feedback array allows the system to encode object information using spatial vibration patterns, as supported by prior research on symbol encoding through haptics [29]. Meanwhile, the linear 1 × 4 arrangement of the distance array offers a simple and intuitive mapping for object proximity. The distance feedback system operates continuously, providing real-time updates on object proximity. The object recognition feedback, conveyed through the 3 × 3 vibration motor array, takes approximately 1.2 seconds per object to complete a full pattern. This timing ensures that users receive detailed haptic information while maintaining smooth navigation. The vibration motor array positions on the armband are shown in Fig. 5.

3.4.1 Distance feedback

The implementation of distance feedback involves the utilization of four vibration motors operating collectively as a unified vibration unit. To effectively span the maximum distance, it is partitioned into four equidistant segments, with each vibration motor assuming control over a distinct segment. The correlation between the ascertained distance value and the intensity of vibration follows an inverse relationship.

This dictates that as the measured distance diminishes, the vibration intensity proportionally increases. The determination of duty cycles for the vibration motors is contingent upon the obtained distance measurements and is achieved through the application of specific formulations delineated in the subsequent exposition. Equation 1 is used to calculate the duty cycle range based on maximum and minimum duty cycles. Equation 2 is used to calculate a temporary variable to simplify the equations. Equation 3 shows the calculation of the duty cycles of the vibration motors based on the duty cycle range parameter T and the temporary variable M .

$$T = DC_{max} - DC_{min} \quad (1)$$

$$M = (D_{max} - D) * 4 * T / D_{max} \quad (2)$$

$$DC_n = \begin{cases} 0, & \text{if } M \leq (n-1) * T \\ M - (n-1) * T + DC_{min}, & \text{if } (n-1) * T \leq M \leq n * T \\ DC_{max}, & \text{if } M \geq n * T \end{cases} \quad (3)$$

By the provided equations, variations in vibration intensity occur in response to alterations in the measured distance parameter. DC_n represents the duty cycle of the n 'th vibration motor. Complementary static parameters, inclusive of the maximum duty cycle DC_{max} , minimum duty cycle DC_{min} , and maximum distance range D_{max} , further characterize the system. The minimum duty cycle, signifying the initiation point of vibration, is established at 50%, as vibration activation remains inhibited for duty cycle values below this threshold. Conversely, the Maximum Duty Cycle is defined at 100%, with the potential for reduction to enhance user comfort if deemed necessary. Notably, the upper limit of the distance range is fixed at 500 cm, although extensibility to 1200 cm is feasible, albeit at the expense of diminished distance resolution conveyed to the user.

3.4.2 Object feedback

The object feedback methodology employed in this study is rooted in the generation of haptic patterns using a 3x3 vibration motor array. These distinct patterns are designed to correspond to specific object classes, such as individuals, automobiles, and chairs, among others. This methodology was initially proposed by Novich et al. Their research highlighted a critical limitation when using a single motor for conveying information, the inability to convey nuanced distinctions due to uniform vibration intensity. Furthermore, the spatial method alone was deemed insufficient for achieving a comprehensive understanding of these patterns. Through a series of experiments, Novich et al. established that the spatiotemporal method emerged as the most effective approach

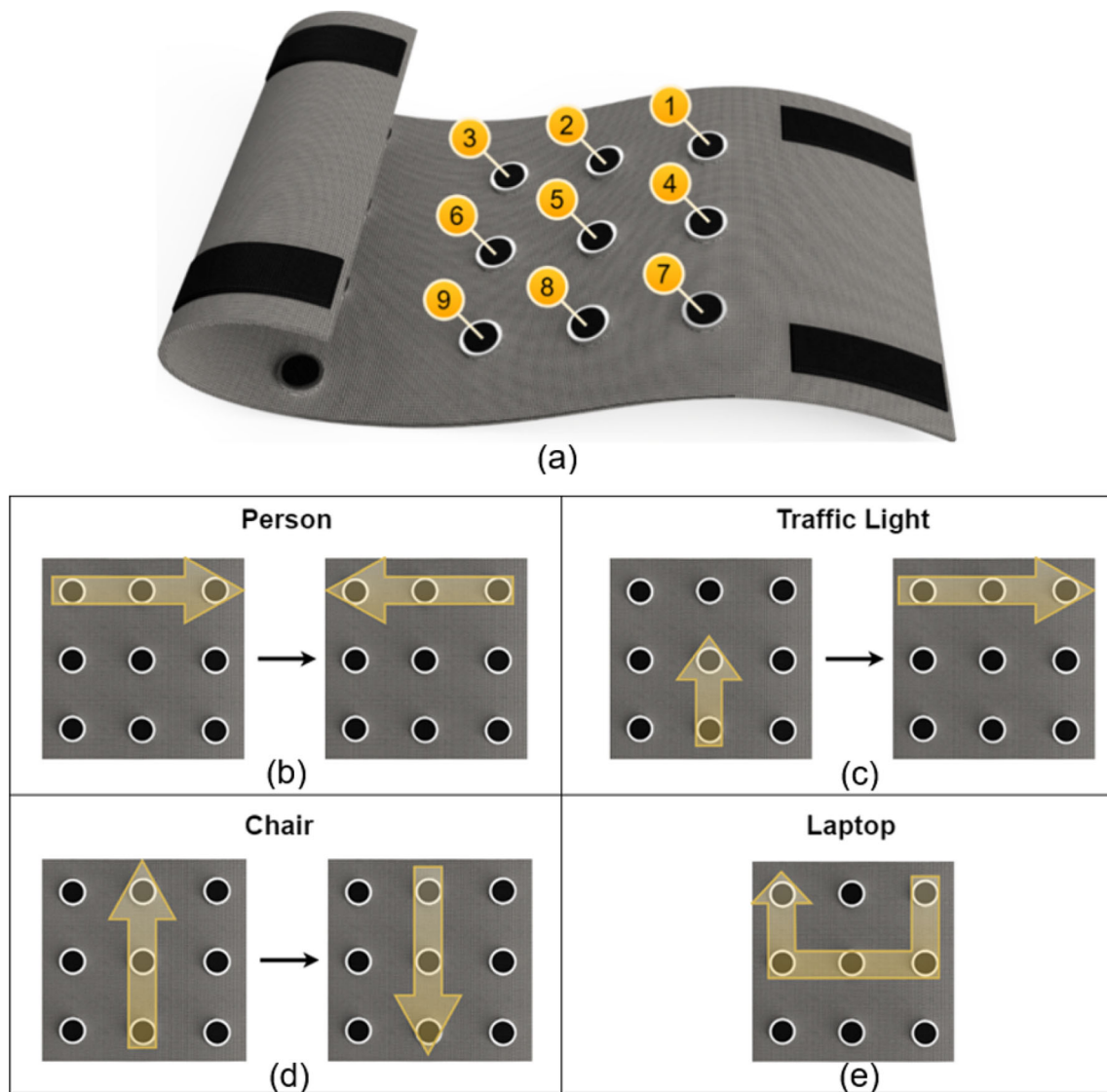


Fig. 6 The illustration of some of the object patterns used in the haptic feedback system. **(a)** The enumeration of the object feedback motors, **(b, c, d, e)** Visual representation of some of the vibration patterns for person, traffic light, chair, and laptop

for enabling users to discern and interpret these patterns. In the spatiotemporal method, the vibration motors work in a specific order [29].

The vibration patterns are inherently contingent upon the user's recognition performance, representing a critical aspect of our investigation. These patterns comprise a sequence of precisely 10 tactile taps, each exhibiting a consistent duration of 120 milliseconds per tap. Consequently, the complete vibration pattern for each distinct class consumes a total temporal interval of 1.2 seconds. Within the temporal framework of each tap, the operational status of the motors in the 3x3 motor array is meticulously controlled. Specifically, at the onset of each tap, a distinct motor is selectively activated, or all motors may be promptly deactivated for a brief delay period. The illustrations of the spatiotemporal vibration pat-

terns for some of the object classes and the numbers of the vibration motors in the array are shown in Fig. 6. Table 2 shows the working order of the motors corresponding to the object classes to generate patterns. This granular control over motor activation within the array underscores the precision and discriminative potential of our tactile feedback system.

4 Results and discussions

The performance of the developed wearable device was assessed through two distinct experiments involving four participants. All participants in the study were university students with an approximate age range of 24–26 years. The participant group consisted of three males and one female.

Table 2 The object patterns for object classes. Each pattern contains 10 Taps to generate unique vibration patterns

Object Class	Patterns Taps									
Person	1	1	2	2	3	3	3	2	2	1
Bicycle	2	0	2	0	2	0	0	0	0	8
Car	1	0	1	0	1	0	0	8	8	8
Motorcycle	2	0	2	0	2	0	0	8	8	8
Bus	6	0	6	0	6	0	0	8	8	8
Train	6	6	0	6	0	0	8	8	8	8
Truck	6	6	0	6	6	6	0	8	8	8
Traffic light	8	8	0	5	5	0	0	1	2	3
Fire hydrant	8	8	0	5	5	0	0	3	2	1
Bench	3	0	3	3	3	0	4	0	4	0
Backpack	3	0	0	0	5	5	0	0	0	3
Bottle	8	0	0	8	0	0	0	0	0	5
Cup	8	0	0	5	0	0	0	0	0	2
Chair	2	2	5	5	8	8	8	5	5	2
Bed	2	2	2	2	6	6	6	3	3	3
Dining table	6	6	0	1	1	0	6	3	3	3
Tv	8	8	0	6	0	6	0	6	0	6
Laptop	3	3	0	6	5	4	0	0	1	1
Keyboard	1	5	3	5	1	5	3	5	1	5

In each experiment, every participant executed five repetitions, during which the timing and walking path were meticulously recorded. In the experimental setup, users wear the device as shown in Fig. 7. It is important to note that our experiments were conducted with blindfolded sighted participants rather than individuals who are blind or visually impaired (BVI). While we assumed that their task completion times and behaviors would be similar to BVI individuals, this assumption remains untested. In reality, visually impaired individuals may have more refined spatial awareness and adaptive skills, which could influence their navigation efficiency. Future studies should include participants with varying levels of visual impairment to better understand how the device performs in real-world scenarios.

In this section, we present the results and discuss the outcomes of the experiments involving a cohort of four participants. The initial phase of our study involved training these participants in recognizing vibration patterns corresponding to 19 distinct classes of objects. Impressively, our training regimen, spanning four hours, culminated in a remarkable 92% recognition accuracy. Following this training, we proceeded to assess the overall performance of the device. This assessment encompassed two distinct tasks carried out by the participants while blindfolded in the 40 m² indoor area. The first task involved locating a chair within an empty room, while the second task entailed identifying a person within a room obstructed by obstacles. The subsequent

**Fig. 7** Front, back, and side view of a blindfolded participant wearing the Vis-Assist device. The controller unit is carried in a backpack

sections will delve into the detailed outcomes of these experiments and their implications for the efficacy of the proposed device.

This study was conducted in accordance with ethical guidelines, and ethical approval was obtained from the Ethical Committee of the University where the project is developed. All participants were informed about the study's objectives and provided written consent before participating. The details of the Ethics Approval are provided in the Statements and Declarations section.

4.1 Participant training

A specialized training mode has been developed within the software to enable individuals to be trained. In this training mode, the processes of object recognition and classification are intentionally bypassed. The device is worn by the individual, and multiple repetitions of each vibration pattern are administered, accompanied by informative feedback regarding the object class associated with each pattern. After that, they were tested by applying random object patterns. This approach ensures that users are equipped with the knowledge to discern and interpret object classifications based on the vibrational cues of the device. Then, the confusion matrix is created according to the results, which is shown in Fig. 8. In the confusion matrix, coloring has been applied separately because the occurrence of each class is not the same in random pattern generation.

4.2 Experiments

The experimental participants wear the device and use a band to be blindfolded, experiencing the assistance ability of the wearable device. Subsequently, these individuals engage in an active exploration of their immediate surround-

	person	bicycle	car	motorcycle	bus	train	truck	traffic light	fire hydrant	bench	backpack	bottle	cup	chair	bed	dining table	tv	laptop	keyboard
person	7	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	3	0
bicycle	0	11	1	3	1	0	0	0	0	0	1	2	0	0	0	0	0	0	0
car	0	0	14	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
motorcycle	0	0	0	7	2	0	0	0	0	0	0	0	0	1	0	0	0	0	0
bus	0	0	0	0	16	2	0	0	0	0	0	0	0	0	0	0	0	0	0
train	1	0	0	0	3	5	0	0	0	0	0	0	0	0	0	2	0	1	0
truck	0	0	0	0	2	0	9	0	0	0	0	0	0	1	1	0	0	1	0
traffic light	0	0	0	0	0	0	0	10	1	0	1	0	0	0	0	0	0	0	0
fire hydrant	0	0	0	0	0	0	0	2	7	0	0	0	2	0	1	0	0	0	0
bench	0	0	0	0	0	1	2	0	0	4	0	0	0	0	0	0	0	0	0
backpack	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0	1	0	0	0
bottle	0	1	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0
cup	0	0	0	0	0	0	0	0	1	0	0	4	21	0	0	0	0	0	0
chair	0	0	0	0	0	0	0	0	0	0	0	0	0	23	0	1	0	1	0
bed	1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	1	0
dining table	0	0	0	0	0	1	0	2	0	0	1	0	0	0	0	6	0	3	0
tv	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	6	1	1
laptop	0	0	0	0	1	1	0	0	0	3	0	0	0	0	0	1	0	11	1
keyboard	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	13

Fig. 8 Confusion matrix depicting the recognition performance of four experiment participants. These subjects received approximately four hours of training to identify various object patterns. The test involved recognizing randomly presented object patterns. Coloring was made separately for different object classes

ings through controlled head movements. When the device detects and classifies an object within the designated region of interest, it proceeds to initiate sequential feedback to the user. This protocol entails the delivery of a specific vibrational pattern tailored to the detected object, followed by distance feedback vibrations. Conversely, in cases where no objects are detected within the defined region of interest, the device exclusively administers distance feedback vibrations. The aligning scenarios of the region of interest and the detected object are shown in Fig. 9.

4.2.1 Experiment 1: finding chair

The first task in the experiments is the localization of a chair within an empty room. Each experimental trial commences with the random placement of both the user and the chair within the designated room. Subsequently, the user embarks on the task of locating the chair. To achieve random placement, the experiment spectator first arranged the positions of the chair and the participant, without informing the participant about the layout. Once the room setup was complete,

the blindfolded participant was brought in and placed at the predetermined position. Throughout each experimental iteration, meticulous records are maintained, capturing both the trajectory of the user's movements and the corresponding completion times. Notably, concurrent with the user's task execution, real-time monitoring of the device's functionality is facilitated by a remote computer system. This remote computer establishes a connection with the Jetson Nano platform employing the RDP (Remote Desktop Protocol) communication protocol, ensuring seamless monitoring and data collection during the experimental trials. The experimental area of the Finding Chair Experiment is shown in Fig. 10. The positions of the chairs, the starting positions of the participants, and their paths for each trial are shown in Fig. 11. Participant trajectory analysis reveals that Participant 1 consistently demonstrated clear and direct paths across all five trials. In contrast, the trajectories of the other participants exhibited circular patterns, suggesting moments of uncertainty or confusion during navigation.

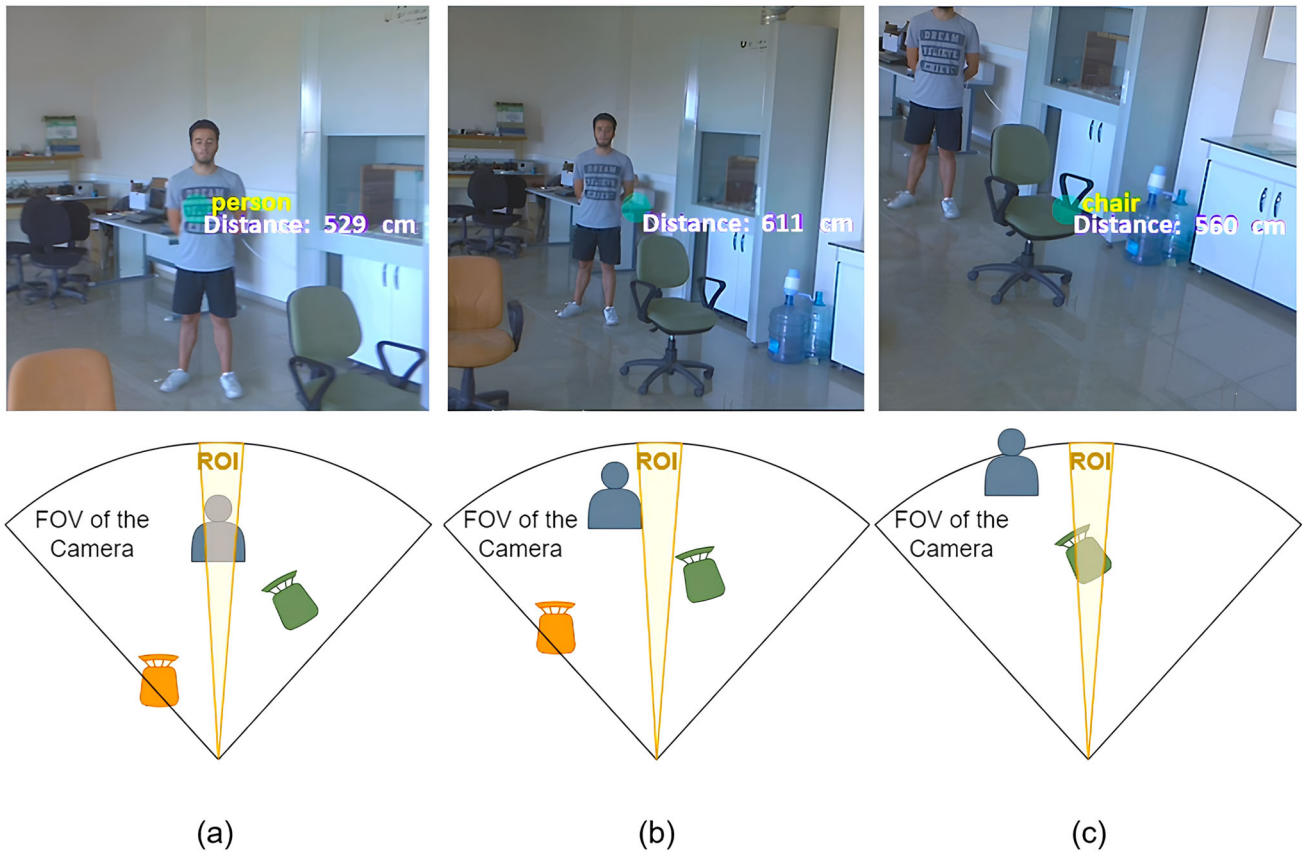


Fig. 9 Illustration of object detection alignment within the region of interest (ROI). (a) A person is accurately detected within the ROI, (b) The ROI is devoid of any detectable objects, (c) A chair is identified within the ROI. Each scenario also displays the distance of the detected object from the camera

Fig. 10 Experimental area of the Finding Chair Experiment. Inset represents the real-time detection of the chair using Vis-Assist device with corresponding distance measurement



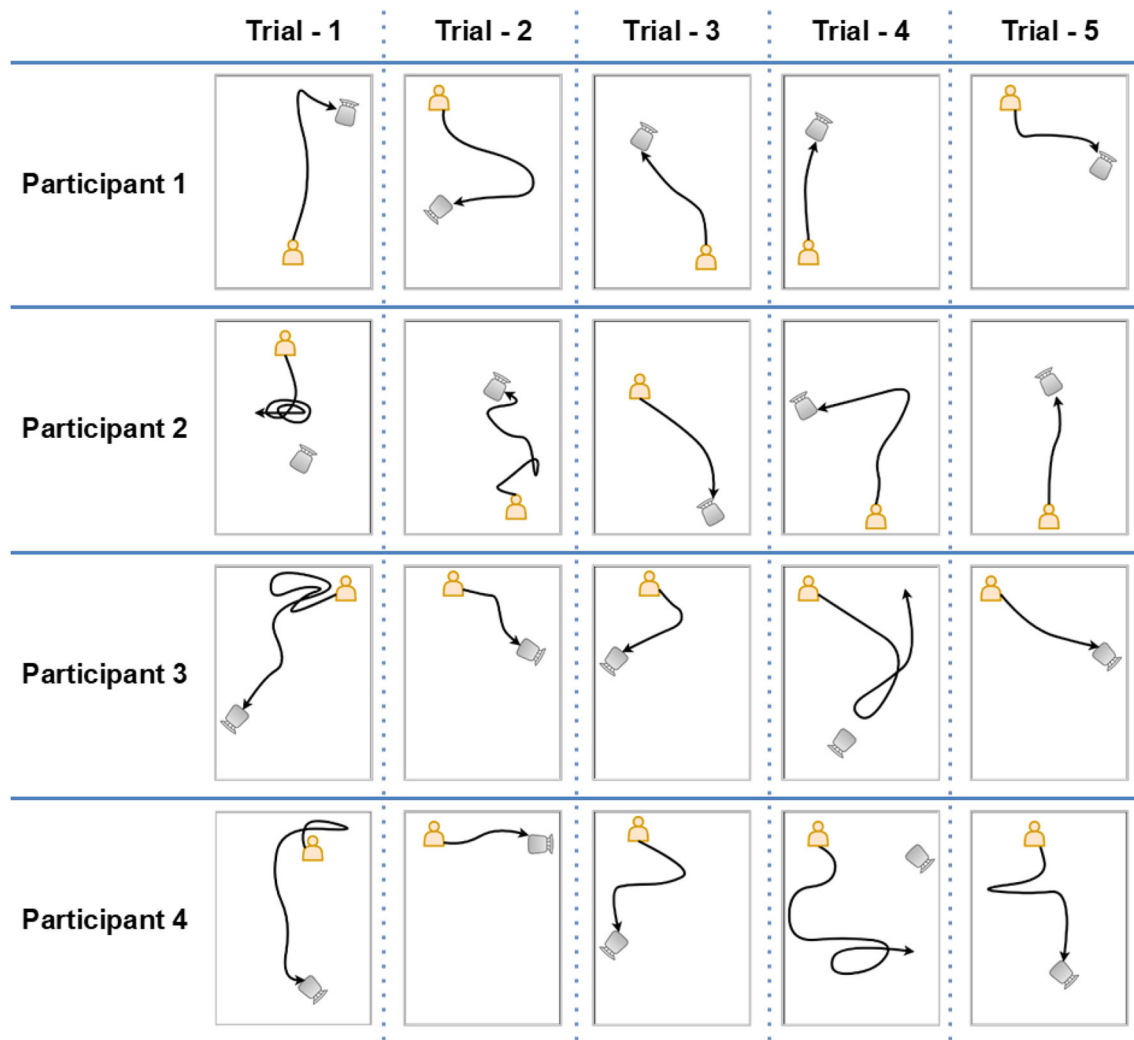


Fig. 11 The randomly placed chairs and the participants' paths during the Finding Chair Experiment trials

4.2.2 Experiment 2: finding person with obstacles

The second experiment focuses on locating a person in a closed room containing randomly placed obstacles, particularly chairs. The experimental area, including the positions of the person and the chairs used as obstacles, was designed by the experiment spectator before each trial. Then, the blindfolded participant was guided into the room and placed at a randomly selected starting point without prior knowledge of the layout. Same four participants take part in this experiment, each conducting the procedure five times, resulting in a total of 20 trials. The goal is to find the person amidst the obstacles. During each trial, the experiment records both the total time taken and the path chosen by the participants to reach the target person. The primary challenge involves maintaining the correct direction to reach the target while navigating around the obstacles. To overcome this challenge, participants employ a specific methodology. Initially,

they scan the entire environment by moving their heads in all directions to locate the target person. Subsequently, they move in that direction while periodically checking for nearby obstacles by looking slightly downward. After a certain period, they reassess the target direction, recognizing that participants may change direction while avoiding obstacles. This approach proves successful in achieving the experiment's objective.

Initially, participants experienced apprehension about colliding with objects. However, after a few trials, they gained confidence in the device's performance and conducted subsequent experiments with a sense of security. Figure 12 shows the experimental area of the Finding Person with Obstacles Experiment. Fig. 13 shows the positions of the chair and the target person, the starting position of the participants, and their paths for each trial. The same four participants took part in both experiments by matching participant number, ensuring consistency in evaluating the system's effectiveness.

Fig. 12 Experimental area of the Finding Person with Obstacles Experiment. Inset represents the real-time detection of the person using Vis-Assist device with corresponding distance measurement

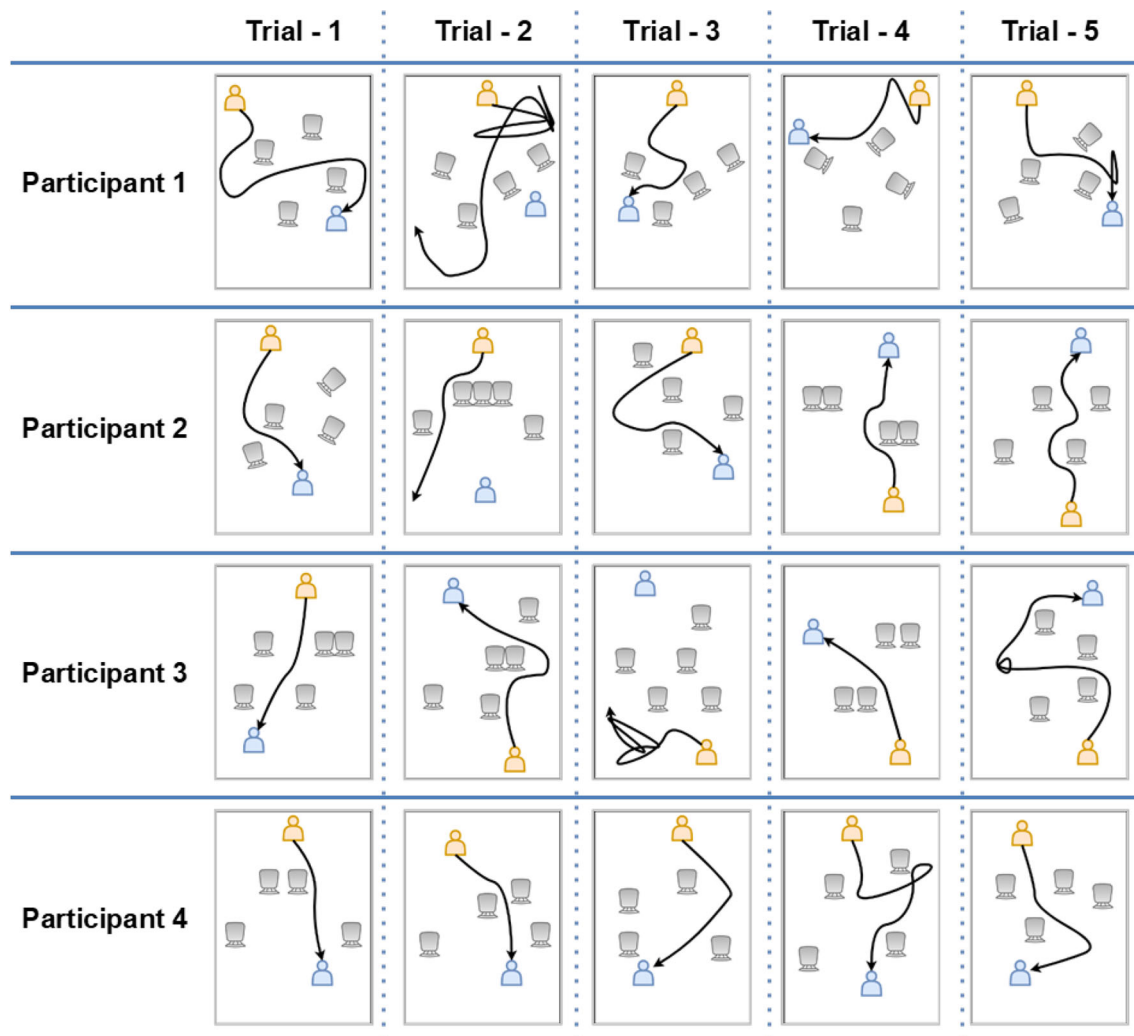
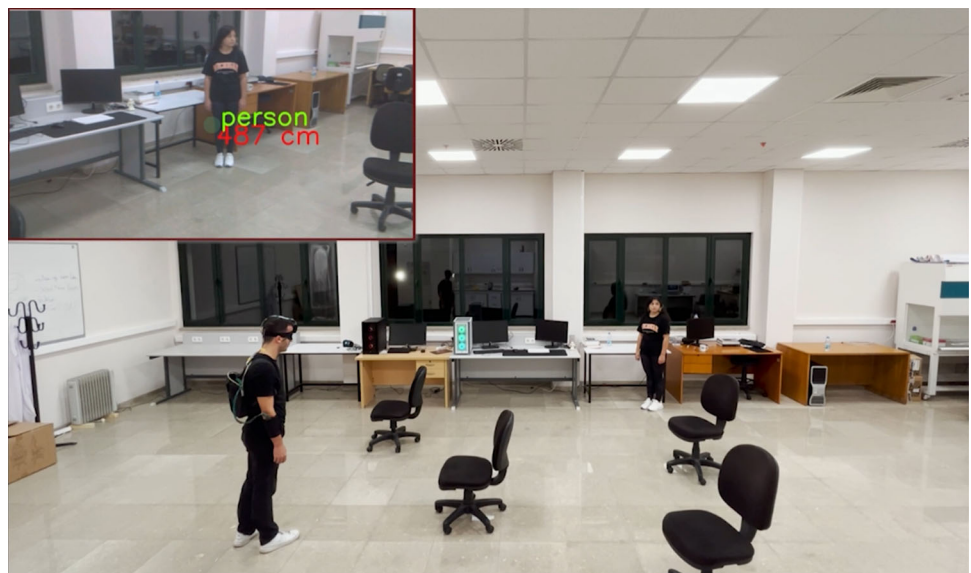


Fig. 13 The positions of the randomly placed chairs and the target person, and the participants' paths during the Finding Person with Obstacles Experiment trials

Table 3 The results of the Finding Chair and Finding Person Experiments.

Experiment	Finding Chair (without obstacles)	Finding Person (with obstacles)
Number of Subjects	4	4
Total Number of Trials	20	20
Success Rate (%)	90	85
Average Time (seconds)	94.4	121.5

This approach eliminates variability in participant experience and allows for a direct comparison of performance across different experimental conditions. Participant trajectories indicate a correlation between participant behavior and path patterns. For instance, Participant 2 exhibited predominantly linear paths, while Participants 1 and 3 demonstrated circular path segments indicative of uncertainty or confusion during specific trials. Optimal participant performance in this experiment would involve exhaustive environmental scanning before initiating movement toward the target. Additionally, the spatial distribution of chair obstacles significantly influenced task success.

4.3 Discussions

The experimental findings herein demonstrate the efficacy of the devised apparatus in facilitating autonomous navigation for individuals with visual impairments. The system empowers users to independently survey their surroundings, identifying objects and gauging their respective distances. The conducted experiments reveal that, on average, users are capable of locating a predefined object, exemplified by a chair, within a 40 m^2 vacant space in under 94 seconds. Moreover, users are proficient in locating objects while circumventing obstacles within the same environment, taking an average of less than 121 seconds to do so. Since all participants shared a similar educational background and age range, there were no significant demographic differences that could cause performance variations in experiments. Differences in navigation performance may be attributed to individual variations in adaptation to the haptic feedback system. The proposed device differentiates itself from existing literature by providing object class information without auditory output. To the best of our knowledge, no other wearable device designed for visually impaired individuals offers such a comprehensive range of information by using haptic feedback. Moreover, the system operates independently without requiring an external server, as all processing is conducted on the edge device. The low-cost components contribute to the affordability of the device, with an approximate total cost of 150 USD. The main cost comes from the Jetson Nano, priced at around 100 USD, followed by the rechargeable

battery at approximately 30 USD. The remaining components do not contribute significantly to the overall cost. These results underscore the potential of the proposed system to significantly enhance the quality of life for visually impaired individuals by affording them greater autonomy and safety in their daily lives. The timing and the success rate results are shown in Table 3 for each experiment. Also, The experiment methodology is shown in a video which is available in the supplementary materials.

It is important to clarify that this study does not compare our system with traditional mobility aids such as white canes. Our focus was on evaluating the feasibility of using haptic feedback for navigation and object detection. During the experiments, blindfolded participants first determined the direction of the target before moving toward it while avoiding obstacles. This initial direction-finding process might be more challenging with a cane, which primarily provides tactile feedback about immediate surroundings rather than distant objects. However, we do not suggest that our system is a replacement for mobility aids. Instead, it should be viewed as a complementary technology that could assist visually impaired individuals in daily activities by providing additional environmental awareness.

4.4 Future perspectives

In this study, fully sighted participants were used as surrogates for BVI individuals and were blindfolded during the experiments. This approach was chosen to evaluate the feasibility of the system, but we acknowledge that individuals with long-term visual impairment may have different sensory adaptations. Future studies should involve participants with varying levels of visual impairment to further validate the system's effectiveness. Additionally, participant demographic details, including education level and profession, could be recorded to better contextualize the training results.

Our study highlights the potential benefits of haptic feedback in specific scenarios where audio-based solutions may be less practical. Rather than presenting haptic feedback as an alternative, we view it as a supportive modality that can complement audio feedback to enhance accessibility. Future developments could explore hybrid systems that integrate both feedback methods, offering users a more flexible and adaptable experience.

In this proof-of-concept study, the results indicate that while training for 19 object classes initially took 4 hours, this duration could decrease as users become more familiar with the system. Moreover, a visually impaired user may find this training period reasonable given the potential long-term benefits for daily life. Future improvements, such as increasing the number of vibration motors or refining the feedback system, may further reduce training time and enhance the overall learning experience.

The participants in the experiment expressed that they were comfortable with the wearability of the device and reported no discomfort under the experimental conditions. Although the device has been developed as a proof of concept to demonstrate the feasibility of haptic-based assistive navigation, future work will focus on optimizing the design, reducing its size, and enhancing wearability to make it more suitable for everyday public use. Considering the size of the electronic components used, there is ample room for miniaturization of the device. In addition, involving visually impaired individuals in the design process could help ensure that the device meets both their functional and aesthetic needs, improving acceptance and comfort in real-world scenarios.

In future perspectives, several enhancements hold promise for the wearable device under consideration. Integration with a dedicated mobile application presents an avenue for tailoring device settings to individual preferences. These settings may encompass the identification of object classes for detection, establishment of maximum measurement distances, selection of operating modes, and customization of vibration intensity levels. Moreover, the adoption of more robust mobile computing platforms, such as the Jetson ORIN, stands as a viable proposition. This advancement would facilitate the execution of more intricate machine learning algorithms and the incorporation of supplementary techniques like Simultaneous Localization and Mapping (SLAM) to enhance navigational capabilities. While the current design of the haptic feedback armband already offers an expanded channel for information transmission to the user, there is room for further refinement to accommodate additional data delivery for an enriched user experience. In addition to training for object pattern recognition, systematic training to interpret distance information based on vibration intensity can also be conducted. This supplementary training may improve navigation accuracy and boost user confidence in estimating object proximity. The number of object classes can be expanded, albeit at the cost of increased training time for users to familiarize themselves with the haptic pattern dictionary. While the current design of the Vis-Assist device provides effective haptic feedback, future work will focus on improving usability and comfort. This includes refining the haptic feedback mechanism, optimizing motor placement, and exploring wireless communication to eliminate wired components. Additionally, efforts will be made to minimize hardware size and enhance the overall wearability of the device to ensure a seamless user experience in real-world applications.

While this study involved a small number of participants, it serves as an initial proof of concept for the feasibility of haptic-based assistive navigation. We recognize that a larger and more diverse participant group would provide deeper insights into the effectiveness of the system. Future research should include a wider range of users, including individuals

with varying levels of visual impairment, to ensure more comprehensive testing. Additionally, testing the device in more complex real-world environments would help evaluate its adaptability and practicality for daily use.

Also, we plan to include visually impaired users in the design and testing of the system by following user-centered and participatory design methods. This will help us better understand their needs and improve how the device works for them. Also, while this version of the system provides only haptic feedback, we collect information using both visual and distance sensors. We aim to improve the user experience by combining haptic feedback with audio in future versions. This direction is supported by recent research showing that using more than one type of feedback can make assistive devices easier and more comfortable to use [30].

5 Conclusion

Our study introduces Vis-Assist, a novel wearable device designed to enhance the daily lives of visually impaired individuals by providing real-time environmental feedback. Unlike existing solutions, Vis-Assist integrates a camera and 1D Lidar with the YOLOv7 model to detect and classify objects, delivering haptic feedback through vibration motor arrays. This approach ensures that users receive valuable information without interfering with their natural hearing. While assistive technologies often use audio feedback to provide information, haptic feedback offers an alternative that does not interfere with hearing and can serve as a complementary approach, allowing users to receive environmental information while engaged in other auditory tasks. We conducted two types of experiments with four subjects, totaling 40 trials, demonstrating that the device effectively helps users locate objects and navigate obstacles within a room. The minimal training requirement of just four hours and the device's affordability and independence from external servers further highlight its accessibility and practicality. Our results underscore Vis-Assist's potential to significantly improve autonomy and daily functioning for visually impaired individuals, offering a promising tool for enhancing their quality of life.

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Code Availability To facilitate further research in this field, the complete source code for this study has been made publicly available on GitHub: <https://github.com/miralab-ai/vis-assist>.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

Ethics approval All experiments involving human participants were conducted by the approval of Ethical Committee of Izmir Institute of Technology (No: E-41752998- 050.01.04-2200004030). Informed consent was obtained from all individual participants in this study.

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